

Instructions for preparing the solution script:

- Write your name, ID#, and Section number clearly in the very front page.
- Write all answers sequentially.
- Start answering a question (not the part of the question) from the top of a new page.
- Start working right away. There is no late submission form. If you miss the deadline, you need to use the make-up assignment to cover up the marks.

1. Consider a function $f(x) = x^3 + x^2 - 4x - 4$.

(a) (2 marks) State the exact roots of $f(x)$ and construct two different fixed point functions $g(x)$ such that $f(x) = 0$.

(b) (3 marks) Compute the convergence rate of each fixed point function $g(x)$ obtained in the previous part, and state which root it is converging to or diverging.

2. Consider the following function: $f(x) = xe^x - 1$.

(a) (3 marks) Find solution of $f(x) = 0$ up to 5 iterations using Newton's method starting with $x_0 = 1.5$. Keep up to four significant figures.

(b) (4 marks) Consider the fixed point function, $g(x) = \frac{2x+1}{\sqrt{x+1}}$. Show that to be super-linearly convergent, the root must satisfy $x^* = \frac{-3}{2}$.

3. (a) (3 marks) Consider the function $f(x) = x^2 - x + 1$ and starting point $x_0 = 0$. Show that the sequence using Newton's method $x_1, x_2, \dots$ fails to approach a root of $f(x)$.